

Exercises

Problem 1

A decision maker chooses among monetary lotteries with three possible outcomes: 3500, 2800, and 0.

Choice 1. The decision maker compares:

$$A : \begin{cases} 3500 & \text{with prob. 0.30,} \\ 2800 & \text{with prob. 0.66,} \\ 0 & \text{with prob. 0.04,} \end{cases} \quad B : 3500 \text{ for sure.}$$

In this first choice, the decision maker selects B .

Choice 2. The decision maker then compares:

$$A' : \begin{cases} 3500 & \text{with prob. 0.30,} \\ 2800 & \text{with prob. 0.00,} \\ 0 & \text{with prob. 0.70,} \end{cases} \quad B' : \begin{cases} 3500 & \text{with prob. 0.00,} \\ 2800 & \text{with prob. 0.34,} \\ 0 & \text{with prob. 0.66.} \end{cases}$$

In this second choice, the decision maker selects A' .

Determine whether these choices can be rationalized by preferences that admit an expected utility (vNM) representation, and justify your conclusion.

Answer

By Definition 6.B.5 in *Microeconomic Theory* (Mas-Colell, Whinston, and Green), a preference over lotteries admits a vNM expected utility representation if there exists a utility index $u(\cdot)$ such that for any lottery

$$L = (p_1, \dots, p_N),$$

its utility is given by

$$U(L) = \sum_{i=1}^N p_i u(x_i).$$

Step 1: Lotteries A and B

$$EU(A) = 0.30 u(3500) + 0.66 u(2800) + 0.04 u(0), \quad EU(B) = u(3500).$$

Using affine invariance of vNM utility, normalize

$$u(3500) = 1, \quad u(0) = 0.$$

Then

$$EU(A) = 0.30 + 0.66 u(2800), \quad EU(B) = 1.$$

Since the decision maker chooses B over A (i.e. $B \succ A$), we require

$$1 > 0.30 + 0.66 u(2800).$$

Solving for $u(2800)$:

$$0.70 > 0.66 u(2800) \implies u(2800) < \frac{0.70}{0.66} \approx 1.06.$$

Step 2: Lotteries A' and B'

$$EU(A') = 0.30 u(3500) + 0 \cdot u(2800) + 0.70 u(0), \quad EU(B') = 0 \cdot u(3500) + 0.34 u(2800) + 0.66 u(0).$$

With the same normalization $u(3500) = 1$ and $u(0) = 0$:

$$EU(A') = 0.30, \quad EU(B') = 0.34 u(2800).$$

Since the decision maker chooses A' over B' (i.e. $A' \succ B'$), we require

$$0.30 > 0.34 u(2800).$$

Solving for $u(2800)$:

$$u(2800) < \frac{0.30}{0.34} \approx 0.88.$$

Conclusion

There exists a non-empty set of utility values satisfying both observed choices. In particular, any

$$u(2800) < 0.88$$

satisfies both $B \succ A$ and $A' \succ B'$ under the normalization $u(3500) = 1$ and $u(0) = 0$. Therefore, the choices are consistent with expected utility.

Problem 2

Consider a decision maker who evaluates uncertain career outcomes according to the expected utility criterion. After completing a program of study, the individual may end up working in one of three sectors: the financial sector (A), the movie industry (B), or the car industry (C).

The decision maker can apply to one of several schools, each leading to different career prospects:

- Attending School S_1 guarantees employment in the car industry.
- Attending School S_2 guarantees employment in the financial sector.
- Attending School S_3 leads to employment in the movie industry with probability 0.1 and in the financial sector with probability 0.9.

The decision maker considers School S_2 to be the least preferred option and is indifferent between Schools S_1 and S_3 .

1. Provide a utility representation consistent with these preferences.
2. Suppose a new option, School S_4 , becomes available. After attending School S_4 , the decision maker works in each sector with probability $1/3$. Rank School S_4 relative to the other schools.
3. Access to School S_4 is limited. Instead of applying directly, the decision maker must apply to a lottery that leads to admission to School S_4 with probability $\alpha \in [0, 1]$ and to School S_2 with probability $1 - \alpha$. For which values of α does the decision maker prefer applying directly to School S_1 rather than participating in this lottery?

Answer

First, we build the expected utilities for the three different schools:

$$\begin{aligned}EU(S_1) &= u(C), \\EU(S_2) &= u(A), \\EU(S_3) &= 0.1 u(B) + 0.9 u(A).\end{aligned}$$

We are given that School S_2 is the least preferred option and that the decision maker is indifferent between Schools S_1 and S_3 . Therefore, the preferences must satisfy:

$$EU(S_1) \sim EU(S_3) > EU(S_2).$$

Indifference between S_1 and S_3 implies:

$$u(C) = 0.1 u(B) + 0.9 u(A).$$

Since S_2 is the least preferred option, it follows that $u(A)$ is the lowest utility level. Moreover, because $u(C)$ is obtained as a convex combination that assigns positive probability to B , the preference ordering must be:

$$u(B) > u(C) > u(A).$$

We now apply the vNM normalization:

$$u(B) = 1, \quad u(A) = 0.$$

Using the indifference condition, we obtain:

$$u(C) = 0.1.$$

Ranking School S_4

School S_4 yields each career outcome with equal probability:

$$S_4 = \{A, B, C\}, \quad P = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

Its expected utility is:

$$EU(S_4) = \frac{1}{3}u(A) + \frac{1}{3}u(B) + \frac{1}{3}u(C).$$

Substituting the normalized utilities:

$$\begin{aligned} EU(S_1) &= 0.1, \\ EU(S_2) &= 0, \\ EU(S_3) &= 0.1, \\ EU(S_4) &= \frac{1}{3}(0 + 1 + 0.1) \approx 0.37. \end{aligned}$$

Hence, the ranking of schools is:

$$S_4 > S_3 \sim S_1 > S_2.$$

Lottery over Schools

Consider the compound lottery:

$$L_\alpha = \{S_4, S_2\}, \quad P = (\alpha, 1 - \alpha).$$

By the vNM reduction axiom, its expected utility is:

$$EU(L_\alpha) = \alpha EU(S_4) + (1 - \alpha) EU(S_2).$$

The decision maker is indifferent between applying to School S_1 and participating in the lottery when:

$$EU(S_1) = EU(L_\alpha).$$

Substituting values:

$$0.1 = \alpha(0.37).$$

Solving for α :

$$\alpha \leq 0.27.$$

Problem 3

Consider a decision maker who evaluates risky prospects according to the expected utility criterion. The utility function over outcomes is given by

$$u(x) = x^a,$$

where the parameter satisfies $a < 1$.

The decision maker faces two lotteries defined over the interval $[0, 1]$:

- Lottery X yields an outcome that is continuously and uniformly distributed on $[0, 1]$.
- Lottery Y yields outcome 0 with probability $1/3$ and outcome 1 with probability $2/3$.

1. Compute the expected utility of lottery X .
2. Compute the expected utility of lottery Y .
3. Determine the value a^* for which the decision maker is indifferent between lotteries X and Y .
4. For $a > a^*$, determine which lottery is preferred by the decision maker.

Answer

First, we know that the outcome of lottery X is uniformly distributed on $[0, 1]$. For a uniform distribution on this interval, the density is given by $f(x) = 1$.

$$\mathbb{E}[u(X)] = \mathbb{E}[X^a].$$

Therefore,

$$\mathbb{E}[X^a] = \int_0^1 x^a dx = \left[\frac{x^{a+1}}{a+1} \right]_0^1 = \frac{1}{a+1}.$$

Hence,

$$\mathbb{E}[u(X)] = \frac{1}{a+1}.$$

Now, for the next part we construct the lottery Y :

$$Y = \{0, 1; p = (\frac{1}{3}, \frac{2}{3})\}.$$

With this lottery, we can compute the expected utility:

$$EU(Y) = \frac{1}{3}u(0) + \frac{2}{3}u(1) = \frac{1}{3}(0)^a + \frac{2}{3}(1)^a = \frac{2}{3}.$$

Thus,

$$EU(Y) = \frac{2}{3}.$$

Now they ask for the value a^* such that $Y \sim X$. In other words, they ask to solve a^* from

$$\mathbb{E}[u(X)] = EU(Y).$$

That is,

$$\frac{1}{a+1} = \frac{2}{3}.$$

Solving for a , we obtain

$$a^* = \frac{1}{2}.$$

For the next part, we evaluate the expected utilities as functions of a .

$$\mathbb{E}[u(X)] = \frac{1}{a+1}.$$

If a increases, then the expected utility of lottery X decreases. On the other hand, the expected utility of lottery Y does not depend on a , since

$$EU(Y) = \frac{2}{3}.$$

Therefore, for $a > a^*$,

$$EU(Y) > EU(X),$$

and the decision maker prefers lottery Y .

Problem 4

Consider a decision maker whose preferences over risky prospects are represented by expected utility, with Bernoulli utility function $u(x)$. The decision maker faces the following lottery:

$$L = \{0, 4, 9\} \quad \text{with probabilities} \quad \left(\frac{1}{6}, \frac{1}{2}, \frac{1}{3}\right).$$

Answer the following questions:

1. Provide the definition of the certainty equivalent of L given a utility function u .
2. Compute the certainty equivalent of L when $u(x) = \sqrt{x}$.
3. Compute the coefficient of absolute risk aversion associated with u .
4. Suppose the decision maker has initial wealth $w = 12$ and owns the lottery L . Determine the minimum amount of money the decision maker is willing to accept in order to sell the lottery.

Answer

Definition of the certainty equivalent.

Given a lottery $L \in \mathcal{L}$ and a utility function u representing the preferences of an expected utility individual, the certainty equivalent of L is the amount $c(L, u)$ such that the individual is indifferent between receiving $c(L, u)$ for sure and playing the lottery L .

First, we compute the certainty equivalent for the lottery L given $u(x) = \sqrt{x}$. For this, we first compute the expected utility of L :

$$U(L) = \frac{1}{6}\sqrt{0} + \frac{1}{2}\sqrt{4} + \frac{1}{3}\sqrt{9} = \frac{1}{6} \cdot 0 + \frac{1}{2} \cdot 2 + \frac{1}{3} \cdot 3 = 2.$$

Now we compute $c(L, u)$. By definition,

$$u(c(L, u)) = \sqrt{c} = 2,$$

which implies

$$c = 4.$$

Therefore,

$$c(L, u) = 4.$$

Now we compute the coefficient of absolute risk aversion. We compute the first and second derivatives of the utility function:

$$u'(x) = \frac{1}{2\sqrt{x}}, \quad u''(x) = -\frac{1}{4x^{3/2}}.$$

Thus, the coefficient of absolute risk aversion is

$$A(x) = -\frac{u''(x)}{u'(x)} = \frac{1}{2x}.$$

Finally, to find the minimum price, we note that the individual starts with initial wealth $w = 12$ and receives a payment q if he sells the lottery. The individual is willing to sell the lottery if

$$u(w + q) = \mathbb{E}[u(w + L)].$$

The possible wealth levels after receiving the lottery outcomes are:

$$\sqrt{12 + 0}, \quad \sqrt{12 + 4}, \quad \sqrt{12 + 9}.$$

We compute the expected utility:

$$\mathbb{E}[u(w + L)] = \frac{1}{6}\sqrt{12} + \frac{1}{2}\sqrt{16} + \frac{1}{3}\sqrt{21} \approx 4.1.$$

Therefore, the individual is willing to sell the lottery if

$$u(w + q) \geq 4.1,$$

that is,

$$\sqrt{12 + q} \geq 4.1.$$

Solving for q :

$$12 + q \geq 4.1^2 \quad \implies \quad q \geq 4.1^2 - 12.$$